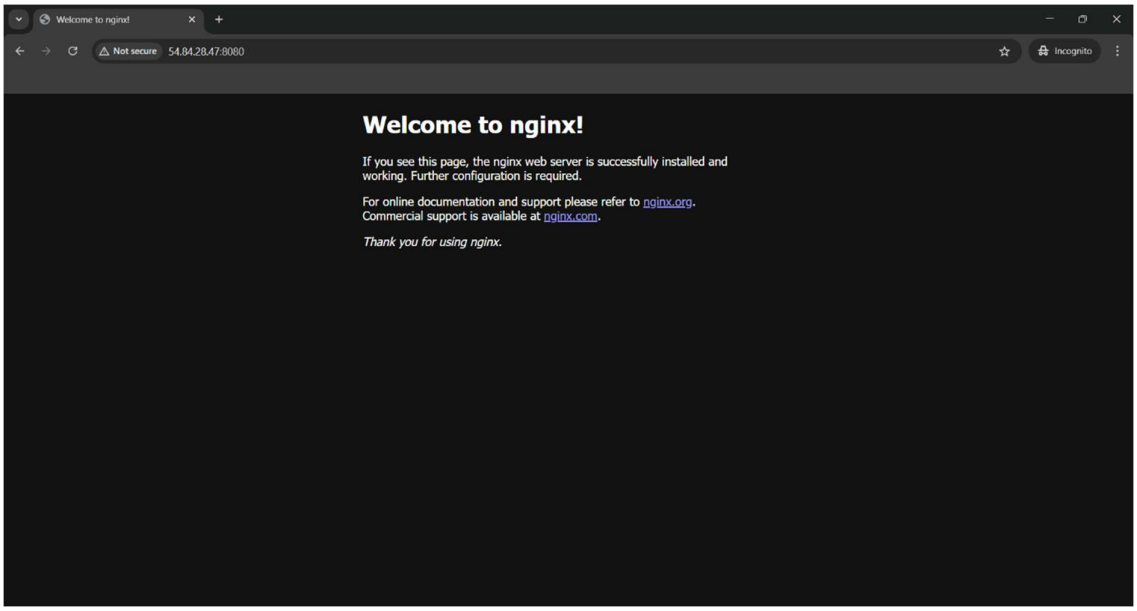
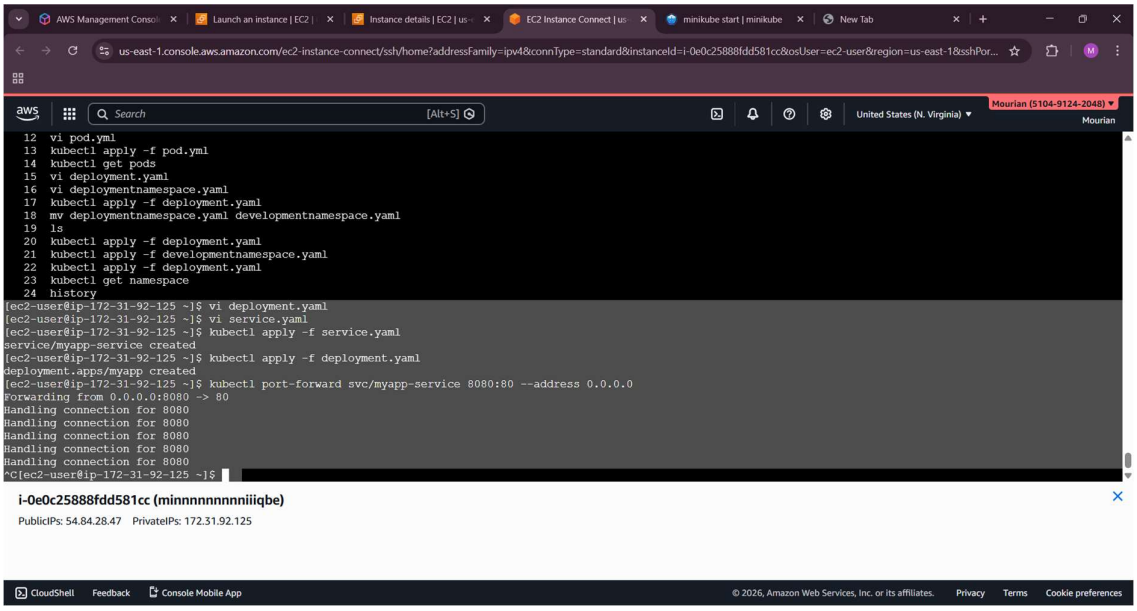
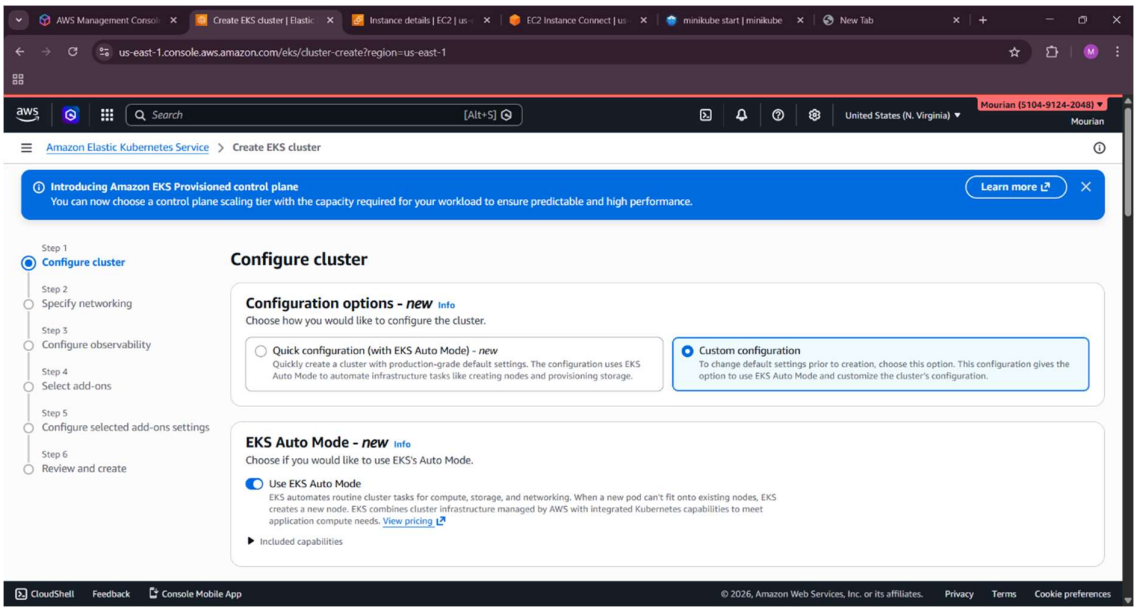
ASSIGNMENT 39

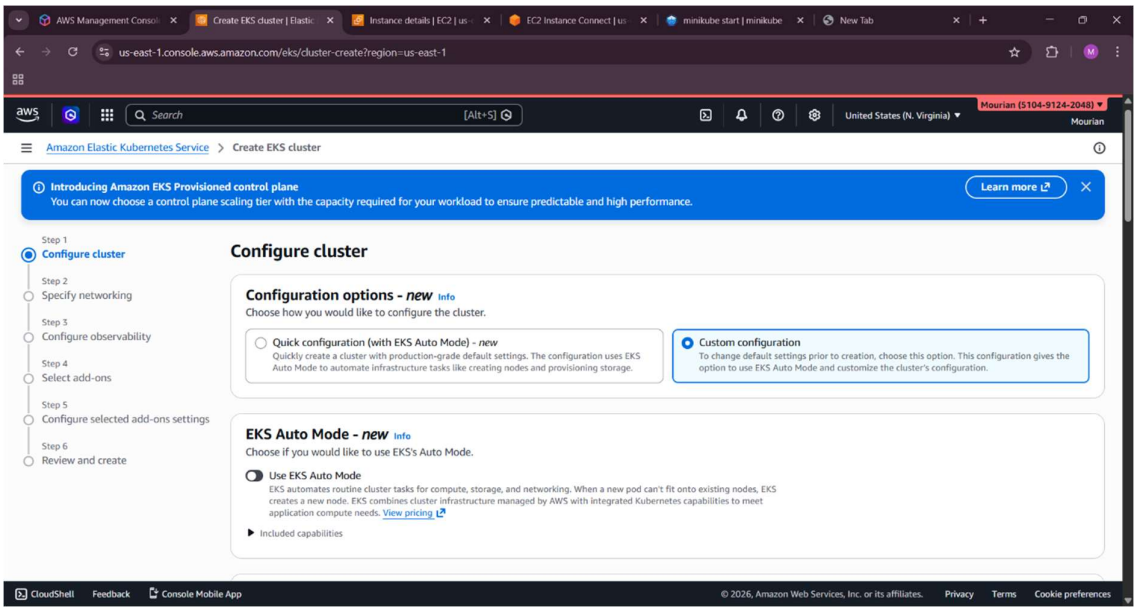
Kubernetes EKS

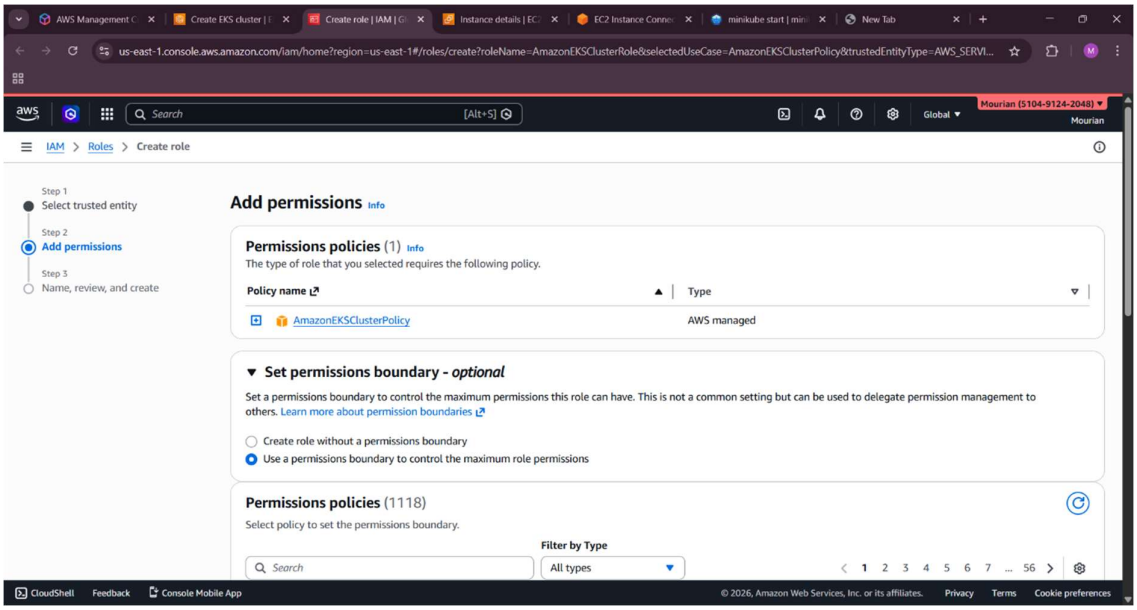




EKS Elastic Kubernetes:






AWS Management C
Create EKS cluster | E
Create role | IAM | Gl
Instance details | EC2
EC2 Instance Conne
minikube start | mini
New Tab
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create?roleName=AmazonEKSClusterRole&selectedUseCase=AmazonEKSClusterPolicy&trustedEntityType=AWS_SERVI...
Search
[Alt+S]
Global
Mourian (5104-9124-2048)
Mourian
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Permissions policies (1) Info
The type of role that you selected requires the following policy.
Policy name
Type
AmazonEKSClusterPolicy
AWS managed
Set permissions boundary - optional
Set a permissions boundary to control the maximum permissions this role can have. This is not a common setting but can be used to delegate permission management to others. Learn more about permission boundaries
Create role without a permissions boundary
Use a permissions boundary to control the maximum role permissions
Permissions policies (1118)
Select policy to set the permissions boundary.
Filter by Type
Search
All types
1 2 3 4 5 6 7 ... 56
CloudShell
Feedback
Console Mobile App
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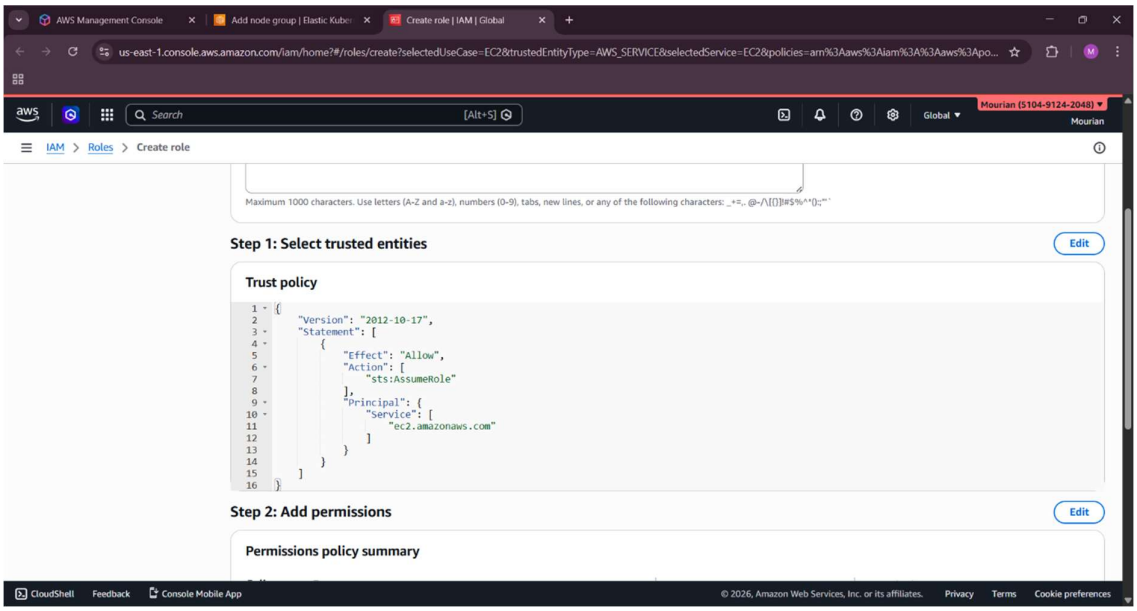
AWS Management Console
Add node group | Elastic Kube
Create role | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?#/roles/create?selectedUseCase=EC2&trustedEntityType=AWS_SERVICE&selectedService=EC2&policies=arm%3Aaws%3Aiam%3A%3Aaws%3Apo...
Search
[Alt+S]
Global
Mourian (5104-9124-2048)
Mourian
IAM > Roles > Create role
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,.@-/\[{}]!#$%^*()&;:"'`
Step 1: Select trusted entities
Edit
Trust policy
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": [
"sts:AssumeRole"
],
"Principal": {
"Service": [
"ec2.amazonaws.com"
]
}
}
]
}
Step 2: Add permissions
Edit
Permissions policy summary
CloudShell
Feedback
Console Mobile App
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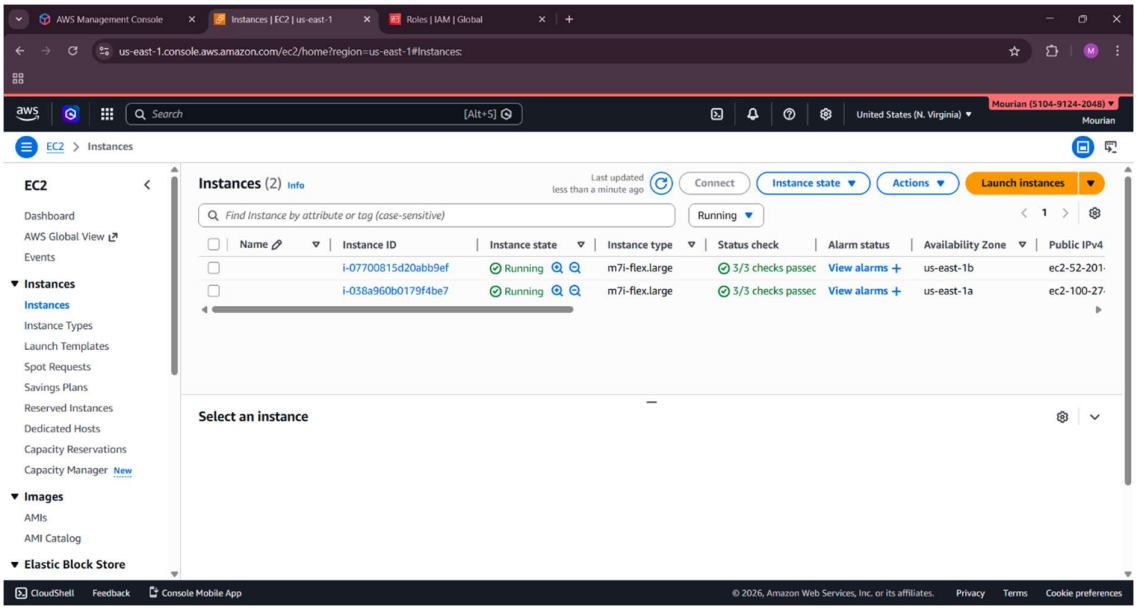
AWS Management Console
Instances | EC2 | us-east-1
Roles | IAM | Global
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search
[Alt+S]
United States (N. Virginia)
Mourian (5104-9124-2048)
Mourian
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances (2) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
Running
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
i-07700815d20abb9ef
Running
m7i-flex.large
3/3 checks passed
View alarms +
us-east-1b
ec2-52-201-
i-038a960b0179f4be7
Running
m7i-flex.large
3/3 checks passed
View alarms +
us-east-1a
ec2-100-27-
Select an instance
CloudShell
Feedback
Console Mobile App
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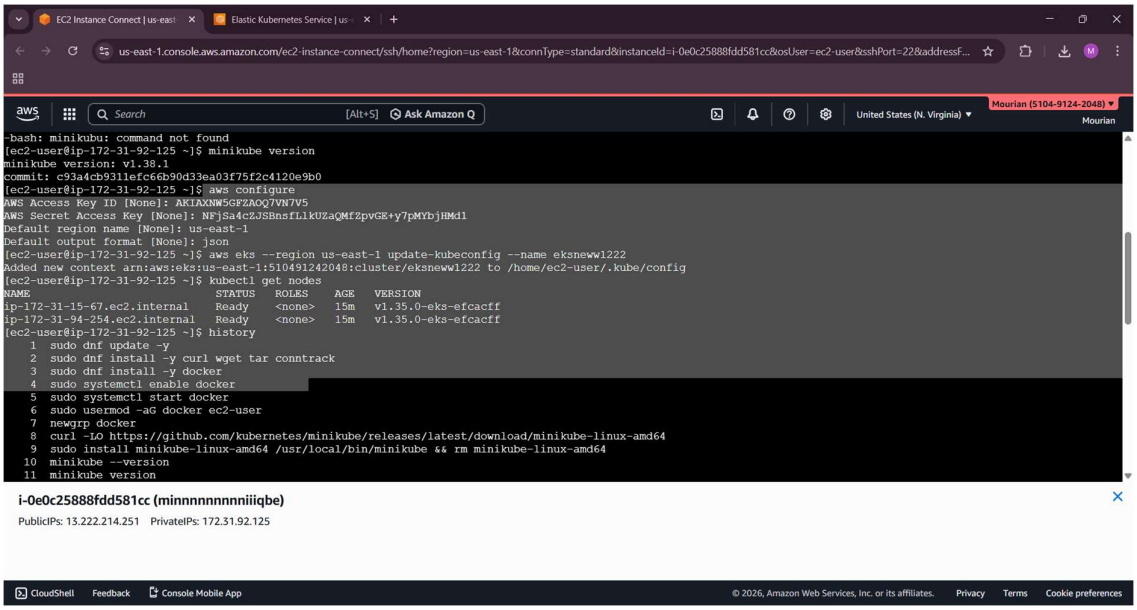
EC2 Instance Connect | us-east
Elastic Kubernetes Service | us-
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0e0c25888fdd581cc&osUser=ec2-user&sshPort=22&addressF...
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Mourian (5104-9124-2048)
Mourian
-bash: minikubu: command not found
[ec2-user@ip-172-31-92-125 ~]$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
[ec2-user@ip-172-31-92-125 ~]$ aws configure
AWS Access Key ID [None]: AKIAXNW5GFZAOQ7VN7V5
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-172-31-92-125 ~]$ aws eks --region us-east-1 update-kubeconfig --name eksneww1222
Added new context arn:aws:eks:us-east-1:510491242048:cluster/eksneww1222 to /home/ec2-user/.kube/config
[ec2-user@ip-172-31-92-125 ~]$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
ip-172-31-15-67.ec2.internal Ready <none> 15m v1.35.0-eks-efcacff
ip-172-31-94-254.ec2.internal Ready <none> 15m v1.35.0-eks-efcacff
[ec2-user@ip-172-31-92-125 ~]$ history
1 sudo dnf update -y
2 sudo dnf install -y curl wget tar conntrack
3 sudo dnf install -y docker
4 sudo systemctl enable docker
5 sudo systemctl start docker
6 sudo usermod -aG docker ec2-user
7 newgrp docker
8 curl -LO https://github.com/kubernetes/minikube/releases/latest/download/minikube-linux-amd64
9 sudo install minikube-linux-amd64 /usr/local/bin/minikube && rm minikube-linux-amd64
10 minikube --version
11 minikube version
i-0e0c25888fdd581cc (minnnnnnnnniiiqbe)
PublicIPs: 13.222.214.251 PrivateIPs: 172.31.92.125
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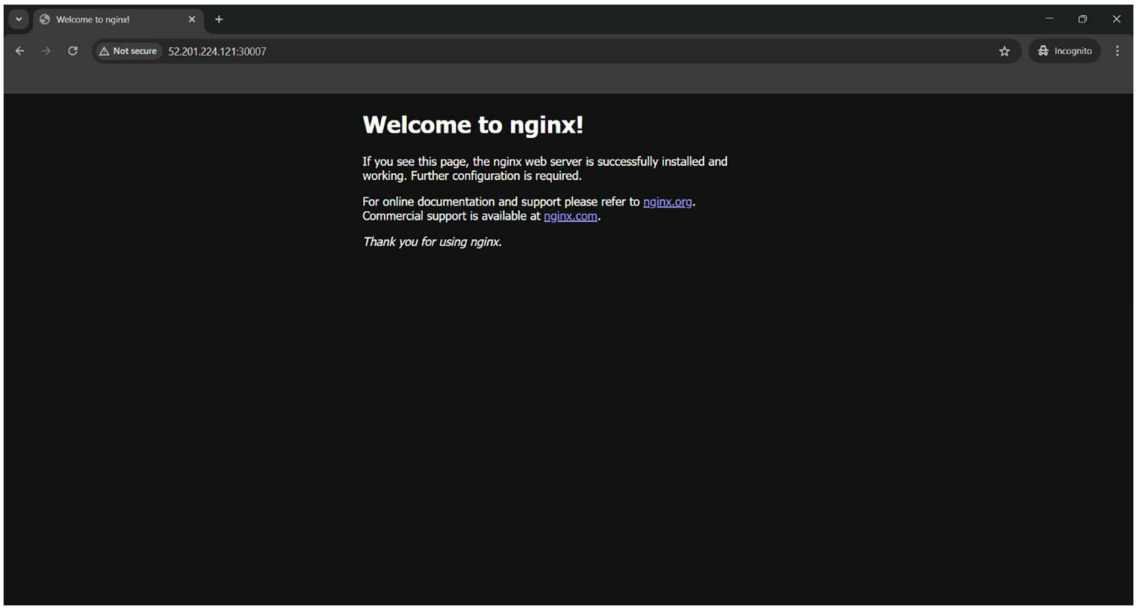
This output from Eks two instance nginx web hosting :



Deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
```

```
        ports:
        - containerPort: 80
```

Service.yaml:

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30007
```